👩‍🏫 Staff Guide · Ag Science 1 · Unit 6 Dairy (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in Mr. Torretta's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: “Dairy Production (Blank Out notes), Dairy Products (Blank Out notes), Darigold video, Cheese presentation (Ms. Gholam), Greek Yogurt, Steps of Ice Cream”)	**Level 2 piece**
Dairy Production (Blank Out, 10 pts)	Notes A + slides (WA rank, breeds, let-down, colostrum)
Dairy Products (Blank Out, 10 pts)	Notes B–C + slides (milk parts, truck, pasteurize, homogenize, shelf life, ice cream)
Darigold video (12) · Cheese presentation (12) · Greek Yogurt (10)	Video guides
Steps of Ice Cream (10 pts)	Ice Cream packet (Labs & Projects)
Class slides	Deck: 21 slides, Words to Know first, hold-up-1-2-3 checks, recap
Class notes (Blank Out)	Guided notes: 16 items + key
Class quiz / notes check	Quiz: 10 questions + key
Vocabulary	10 picture cards (cut apart)
—	Review game (match + order) on the class site

2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no “all of the above”
- Show a real milk carton: find “pasteurized,” “vitamin D,” and the sell-by date
- Sort the vocabulary cards: cow words vs. food words

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the “look for” part

4. What to grade

- Dairy allergies or lactose intolerance: no tasting needed for any dairy item; check with Mr. Torretta before any tasting
- **Notes**: same points as the class Blank Out notes when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver, Room 113 (SLE): changes and IEP questions.
- Facts we could not confirm in his files say “check with Mr. Torretta”.

6. Photo credits (open licenses)

HF in der Rhön auf der Weide.jpg — Verum, CC BY-SA 3.0, Wikimedia Commons · Jersey cow J4.jpg — Jamain, CC BY-SA 3.0, Wikimedia Commons · Melkkarussell.jpg — Gunnar Richter Namenlos.net, CC BY-SA 3.0, Wikimedia Commons · Müller milk tanker in Longnor - geograph.org.uk - 7919111.jpg — TCExplorer, CC BY-SA 2.0, Wikimedia Commons · Somerset-Cheddar.jpg — J.P.Lon, CC BY-SA 3.0, Wikimedia Commons · Vanilla bean ice cream (3086700978).jpg — a.pasquier from bellingham, washington, CC BY-SA 2.0, Wikimedia Commons · Diagrams/pictures from Mr. Torretta's class slides: dairy-herd.png

👩‍🏫 Staff Guide · Ice Cream: Steps and Key Points (Level 2)

Ag Science 1 · Unit 6 · goes with the class “Steps of Ice Cream” (10 pts). Level 2 = about 3rd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Steps card**: Mr. Torretta's “Steps and Key Points of Making Ice Cream” sheet (steps + key points). 8 steps, one per box. **check with Mr. Torretta**: whether there is a hands-on ice cream lab (and his recipe), plus dairy-allergy rules for tasting
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “What do we do first?”
- “Why do we heat it?”
- “Point to the step where air goes in.”

3. Modifications

- Steps are given; student fills only the key point from the bank
- 8 steps in a table instead of a blank page

4. What to grade (10 pts)

- Same 10 points: key points table complete
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

6. Photo credits (open licenses)

Vanilla bean ice cream (3086700978).jpg — a.pasquier from bellingham, washington, CC BY-SA 2.0, Wikimedia Commons

🐄 Dairy Production and Dairy Products

Ag Science 1 · Unit 6 Dairy · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Dairy cows

1. Milk and dairy products are Washington's number _____ farm product. Number 1 is apples.
2. In 50 years, the number of dairy cows went down, but milk per _____ went up.
3. Match each breed to its fact:

Holstein	another dairy breed
Jersey	small, brown, creamy milk
Brown Swiss	black and white, most milk

4. The hormone that lets milk down is _____. Stress can block it.
5. A calf's first milk is _____. It has antibodies.

Word bank: **2** **colostrum** **cow** **oxytocin**

B. What is in milk

6. Milk is about 87% _____.
7. Milk's 3 main nutrients are protein, _____, and riboflavin.
8. The main milk protein is _____. It clumps to make cheese.
9. Which are in milk? (circle 2)

lactose (milk sugar) caffeine fat gasoline

Word bank: **calcium** **casein** **water**

C. Processing and products

10. The truck driver checks milk is under _____ °F.
11. _____ heats milk to kill harmful bacteria.
12. _____ breaks up fat so cream does not rise.
13. Vitamin _____ is added to milk to help the body use calcium.
14. Plain ice cream must have at least _____% fat.

Word bank: **10** **40** **D** **Homogenization** **Pasteurization**

Level 2 version of the gen ed “Dairy Production + Dairy Products (Blank Out notes).”

📖 Dairy: Vocabulary Picture Cards (Level 2)

Name: _____

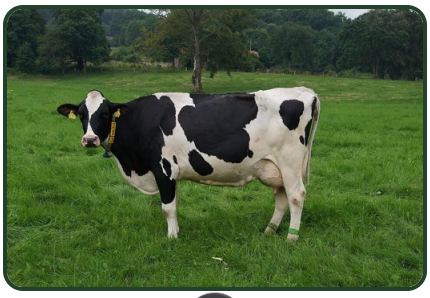
Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



🐄

Dairy cow

A cow raised to give milk. Most U.S. milk comes from cows.



⚫

Holstein

The black-and-white breed; about 80% of registered U.S. dairy cattle. Gives the most milk.



🟤

Jersey

A small brown breed. Its milk is high in fat (creamy).



🥛

Milking parlor

The barn room where cows are milked by machines.



Milk truck

The tank truck that hauls milk from the farm to the plant, kept under 40°F.

Pasteurize

Heat milk to destroy harmful bacteria (pathogens). It does not sterilize milk, so keep it cold.

Calcium

A mineral in milk that builds strong bones and teeth.

Colostrum

The first milk after a calf is born. It has antibodies that protect the calf.



Cheese

A food made when the milk protein (casein) clumps together (coagulates).



Ice cream

A frozen dairy food with at least 10% milk fat. Air whipped in is called overrun.

🍦 Ice Cream: Steps and Key Points

Ag Science 1 · Unit 6 · Level 2



Name: _____

Mr. Torretta's "Steps and Key Points of Making Ice Cream" sheet (steps + key points) · check with Mr. Torretta.

🧺 If there is a tasting or ice-cream-in-a-bag lab (check with Mr. Torretta)

- ☐ 🥛 **1/2 cup** half-and-half or milk
- ☐ 🍬 **1 Tbsp** sugar
- ☐ 🍦 **1/4 tsp** vanilla
- ☐ 🧊 **3 cups** ice + 1/3 cup rock salt (big bag)

🌱 Steps: check each box when you finish

1. 🥣 **Blend**: Mix cream, milk, sugar, and stabilizer. ☐
2. 🔥 **Pasteurize**: Heat the mix. ☐
3. 🌀 **Homogenize**: Break up the fat. ☐
4. 🧊 **Age (cool)**: Cool the mix to about 40°F for hours. ☐
5. 🍫 **Flavor**: Add vanilla, chocolate, or fruit flavor. ☐
6. ❄️ **Freeze + whip**: Freeze while whipping in air. ☐
7. 🍓 **Mix-ins**: Add fruit, nuts, or candy pieces. ☐
8. 📦 **Package + harden**: Fill cartons and freeze very cold. ☐

🍦 Steps and Key Points of Making Ice Cream

#	Step	What happens	Key point (why)
1	🥣 Blend	Mix cream, milk, sugar, and stabilizer	_____
2	🔥 Pasteurize	Heat the mix	_____
3	🌀 Homogenize	Break up the fat	_____
4	🧊 Age (cool)	Cool the mix to about 40°F for hours	_____
5	🍫 Flavor	Add vanilla, chocolate, or fruit flavor	_____
6	❄️ Freeze + whip	Freeze while whipping in air	_____
7	🍓 Mix-ins	Add fruit, nuts, or candy pieces	_____
8	📦 Package + harden	Fill cartons and freeze very cold	_____

Key point bank: **Added after freezing** **Air = overrun** **Hard, stays fresh** **Kills harmful bacteria** **Measure exactly** **Smooth, creamy texture** **Taste** **Whips better**

💡 *Ice cream is at least 10% milk fat. Air whipped in while freezing is called “overrun.” Premium ice cream has more fat and less air.*

🍦 Ice Cream: Steps and Key Points · Prep Sheet

Ag Science 1 · Unit 6 · Level 2



Name: _____

👥 Job roles: circle your job. Initial when it is done.

Job	What you do	Done ✔️
📏 **Measurer**	Measures the milk, sugar, and vanilla.	
🔒 **Sealer**	Seals both bags tight.	
🤲 **Shaker**	Shakes the bag 5–8 minutes (take turns).	
🧽 **Cleaner**	Wipes the table and throws bags away.	

🧰 Tools and supplies: check when you have it

- ☐ 👜 **Small zip bag** for the mix
- ☐ 🛍️ **Big zip bag** for ice + salt
- ☐ 🧤 **Towel or gloves** the bag gets very cold
- ☐ 🥄 **Spoon + cup** to eat

⚠️ Safety and manners focus

- 🥛 Dairy allergy or lactose intolerance? Do not taste. You still get full points.
- 🧤 Rock salt + ice gets very cold. Hold the bag with a towel or gloves.
- 🧼 Wash hands before touching food.
- 🧂 Do not eat the rock salt.

🍦 Ice Cream: Steps and Key Points · Reflection

Ag Science 1 · Unit 6 · Level 2



Name: _____

📸 My photos / proof

☐ Steps table done ☐ Lab (if any) ☐ Clean station

✅ Skills I used today (check all you did)

☐ 🔢 I put all 8 steps in order

☐ 🔑 I matched each step to its key point

☐ 🤝 I did my role

☐ 🧼 I followed the safety rules

⭐ Which step surprised you? Why?

1. _____
2. _____

Ideas: **pasteurize** **aging** **overrun** **hardening**

🔧 Why does ice cream need to be pasteurized?

1. _____
2. _____

Ideas: **kill bacteria** **safe to eat**

✅ Dairy Quiz

Ag Science 1 · Unit 6 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🏆 In Washington, milk is the #2 farm product. #1 is…

A. Wheat | B. Apples | C. Potatoes

2. ⚫ The breed with the most milk is…

A. Jersey | B. Ayrshire | C. Holstein

3. 🧠 Which hormone lets milk down?

A. Oxytocin | B. Insulin | C. Caffeine

4. 🍼 Colostrum is important because it has…

A. Sugar only | B. Salt | C. Antibodies

5. 💧 Milk is mostly…

A. Water | B. Fat | C. Protein

6. 🌡️ Milk should be kept below…

A. 70°F | B. 40°F | C. 100°F

7. 🔥 Pasteurization…

A. Adds fat | B. Kills harmful bacteria | C. Makes milk shelf-stable forever

8. 🌀 Homogenization…

A. Breaks up fat so cream doesn't rise | B. Adds vitamins | C. Freezes milk

9. 🧀 The milk protein that clumps to make cheese is…

A. Lactose | B. Oxytocin | C. Casein

10. 🍦 Plain ice cream must have at least…

A. 10% fat | B. 1% fat | C. 50% fat

🎬 Cheese Presentation (Ms. Gholam)

Ag Science 1 · Unit 6 · Level 2 video guide (modified). Ms. Gholam's presentation in class. Listen for the cheese facts.

Name: _____

While you watch: circle the best answer

1. The milk protein that clumps (coagulates) to make cheese is…

A. Lactose | B. Casein | C. Gluten

2. Cheddar is a semi-hard cheese.

A. False | B. Only in Ferndale | C. True

3. Which is a fresh cheese?

A. Cottage cheese or ricotta | B. Aged parmesan | C. Hard cheddar

After you watch: finish the sentence

4. Two nutrition facts about cheese:

1. _____ 2. _____ _____

5. How many cheese factories are in Ferndale?

_____ _____

6. My favorite cheese is

_____ because _____. _____

Goes with the class “Cheese Presentation Ms. Gholam” (12 pts). Same points when complete.

🎬 Darigold: Nourishing the World

Ag Science 1 · Unit 6 · Level 2 video guide (modified). YouTube, 2 parts: youtube.com/watch?v=DkkuwPJtY6M and youtube.com/watch?v=8RE9-XcxRZA

Name: _____

Before you watch

Where does the milk in your fridge come from?

_____

While you watch: circle the best answer

1. Do milk chocolate M&M's use milk?

A. No | B. Yes | C. Only the blue ones

2. Darigold is owned by…

A. A soda company | B. The government | C. Dairy farm families (a co-op)

3. A rotary milker is…

A. A turning platform where cows are milked | B. A milk truck | C. A cheese machine

4. An anaerobic digester turns manure into…

A. Ice cream | B. Plastic bottles | C. Energy and fertilizer

5. Why do some colleges give athletes chocolate milk?

A. To help them recover after exercise | B. Because it is free candy | C. To help them sleep

After you watch: finish the sentence

6. Milk provides vitamins, minerals, and _____ for growing kids. _____

7. One fact from the video that surprised me:

_____ _____

Goes with the class "Dairigold Nourishing The World Video" (12 pts). Same points when complete.

🎬 Benefits of Greek Yogurt

Ag Science 1 · Unit 6 · Level 2 video guide (modified). Mr. Torretta's Greek Yogurt sheet (class video or reading).

Name: _____

While you watch: circle the best answer

1. Yogurt is…

A. Frozen milk | B. Milk fermented with good bacteria | C. Cheese with fruit

2. The good bacteria in yogurt are called…

A. Antibiotics | B. Vitamins | C. Probiotics

3. Compared to regular yogurt, Greek yogurt has…

A. About twice the protein | B. No protein | C. Less calcium and protein

4. A vitamin found in yogurt:

A. Vitamin Z | B. Caffeine | C. Vitamin B12

After you watch: finish the sentence

5. Two benefits of eating yogurt:

1. _____ 2. _____ _____

6. When buying yogurt, look for

_____ _____

7. The word “yogurt” comes from

_____ _____

Goes with the class “Benefits of Greek Yogurt” (10 pts). Same points when complete.

🐄 Dairy Production and Dairy Products · ANSWER KEY

Ag Science 1 · Unit 6 Dairy · Level 2 guided notes. Use with the class slides.

Name: _____

Fill in each blank with a word from the word bank in that box. Each word is used one time.

A. Dairy cows

1. Milk and dairy products are Washington's number **2** farm product. Number 1 is apples.
2. In 50 years, the number of dairy cows went down, but milk per **cow** went up.
3. Match each breed to its fact:

(Key: Holstein → black and white, most milk · Jersey → small, brown, creamy milk · Brown Swiss → another dairy breed)

Holstein	another dairy breed
Jersey	small, brown, creamy milk
Brown Swiss	black and white, most milk

4. The hormone that lets milk down is **oxytocin**. Stress can block it.
5. A calf's first milk is **colostrum**. It has antibodies.

Word bank: **2** **colostrum** **cow** **oxytocin**

B. What is in milk

6. Milk is about 87% **water**.
7. Milk's 3 main nutrients are protein, **calcium**, and riboflavin.
8. The main milk protein is **casein**. It clumps to make cheese.
9. Which are in milk? (circle 2)

lactose (milk sugar) caffeine **fat** gasoline

Word bank: **calcium** **casein** **water**

C. Processing and products

10. The truck driver checks milk is under **40** °F.

11. **Pasteurization** heats milk to kill harmful bacteria.

12. **Homogenization** breaks up fat so cream does not rise.

13. Vitamin **D** is added to milk to help the body use calcium.

14. Plain ice cream must have at least **10** % fat.

Word bank: **10** **40** **D** **Homogenization** **Pasteurization**

Level 2 version of the gen ed “Dairy Production + Dairy Products (Blank Out notes).”

✅ Dairy Quiz · ANSWER KEY

Ag Science 1 · Unit 6 · Level 2 (modified). 10 questions · 3 choices each · 1 point each.

Name: _____

Circle the best answer. You may have the questions read aloud.

1. 🏆 In Washington, milk is the #2 farm product. #1 is… Production

A. Wheat	**B. Apples**	C. Potatoes

2. ⚫ The breed with the most milk is… Production

A. Jersey	B. Ayrshire	**C. Holstein**

3. 🧠 Which hormone lets milk down? Production

A. Oxytocin	B. Insulin	C. Caffeine

4. 🍼 Colostrum is important because it has… Production

A. Sugar only	B. Salt	**C. Antibodies**

5. 💧 Milk is mostly… Products

A. Water	B. Fat	C. Protein

6. 🌡️ Milk should be kept below… Products

A. 70°F	**B. 40°F**	C. 100°F

7. 🔥 Pasteurization… Products

A. Adds fat	**B. Kills harmful bacteria**	C. Makes milk shelf-stable forever

8. 🌀 Homogenization… Products

A. Breaks up fat so cream doesn't rise	B. Adds vitamins	C. Freezes milk

9. 🧀 The milk protein that clumps to make cheese is… Cheese

A. Lactose	B. Oxytocin	**C. Casein**

10. 🍦 Plain ice cream must have at least… Products

A. 10% fat	B. 1% fat	C. 50% fat

Answers

1-B · 2-C · 3-A · 4-C · 5-A · 6-B · 7-B · 8-A · 9-C · 10-A

Reworded from the gen ed "Dairy Production and Dairy Products (Blank Out notes)". Score: __ / 10.

🎬 Cheese Presentation (Ms. Gholam) · ANSWER KEY

Ag Science 1 · Unit 6 · Level 2 video guide (modified). Ms. Gholam's presentation in class. Listen for the cheese facts.

Name: _____

While you watch: circle the best answer

1. The milk protein that clumps (coagulates) to make cheese is…

A. Lactose | **B. Casein** | C. Gluten

2. Cheddar is a semi-hard cheese.

A. False | B. Only in Ferndale | **C. True**

3. Which is a fresh cheese?

A. Cottage cheese or ricotta | B. Aged parmesan | C. Hard cheddar

After you watch: finish the sentence

4. Two nutrition facts about cheese:

1. _____ 2. _____ **protein; calcium (others OK)**

5. How many cheese factories are in Ferndale?

_____ **check with Mr. Torretta**

6. My favorite cheese is

_____ because _____. _____

Goes with the class "Cheese Presentation Ms. Gholam" (12 pts). Same points when complete. Numbers from the talk (years of cheese making, types of cheese, pounds on pizza) depend on her slides; check with Mr. Torretta.

🎬 Darigold: Nourishing the World · ANSWER KEY

Ag Science 1 · Unit 6 · Level 2 video guide (modified). YouTube, 2 parts: youtube.com/watch?v=DkkuwPJtY6M and youtube.com/watch?v=8RE9-XcxRZA

Name: _____

Before you watch

Where does the milk in your fridge come from?

_____

While you watch: circle the best answer

1. Do milk chocolate M&M's use milk?

A. No | **B. Yes** | C. Only the blue ones

2. Darigold is owned by…

A. A soda company | B. The government | **C. Dairy farm families (a co-op)**

3. A rotary milker is…

A. A turning platform where cows are milked | B. A milk truck | C. A cheese machine

4. An anaerobic digester turns manure into…

A. Ice cream | B. Plastic bottles | **C. Energy and fertilizer**

5. Why do some colleges give athletes chocolate milk?

A. To help them recover after exercise | B. Because it is free candy | C. To help them sleep

After you watch: finish the sentence

6. Milk provides vitamins, minerals, and _____ for growing kids. **protein**

7. One fact from the video that surprised me:

_____ _____

Goes with the class "Dairigold Nourishing The World Video" (12 pts). Same points when complete. Video-specific numbers (cows milked per hour, etc.) are not on this guide; check with Mr. Torretta for his key.

🎬 Benefits of Greek Yogurt · ANSWER KEY

Ag Science 1 · Unit 6 · Level 2 video guide (modified). Mr. Torretta's Greek Yogurt sheet (class video or reading).

Name: _____

While you watch: circle the best answer

1. Yogurt is…

A. Frozen milk | **B. Milk fermented with good bacteria** | C. Cheese with fruit

2. The good bacteria in yogurt are called…

A. Antibiotics | B. Vitamins | **C. Probiotics**

3. Compared to regular yogurt, Greek yogurt has…

A. About twice the protein | B. No protein | C. Less calcium and protein

4. A vitamin found in yogurt:

A. Vitamin Z | B. Caffeine | **C. Vitamin B12**

After you watch: finish the sentence

5. Two benefits of eating yogurt:

1. _____ 2. _____ **protein, calcium/bones, probiotics/gut health**

6. When buying yogurt, look for

_____ **low added sugar; “live and active cultures”**

7. The word “yogurt” comes from

_____ **Turkish (yoğurt) — check with Mr. Torretta**

Goes with the class “Benefits of Greek Yogurt” (10 pts). Same points when complete.